

A Fast Clustering Method for Identifying Rock Discontinuity Sets

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Abstract

Identifying rock discontinuity sets is a major factor for analyses of hydraulic properties and rock mass stability. To perform efficient and precise grouping of discontinuities, a new approach based on clustering by fast searches and finding density peaks is proposed for the identification of rock discontinuity sets. By measuring the similarity of each pair of discontinuities, the local density and controlled distance of each discontinuity can be calculated for a certain cutoff distance. The number of potential clusters and central discontinuities of corresponding clusters can be found by observing the decision graph constructed based on the decision values of all discontinuities in descending order. The discontinuities of each cluster are divided into core discontinuities and outlier discontinuities based on the corresponding boundary density. This strategy can avoid interference from subjective factors and improve the accuracy of the clustering analysis. The new approach is verified using artificial data, and the appropriate cutoff distance thresholds for identifying rock discontinuity sets are given. Finally, the new approach is applied to group discontinuities in an actual underground mine. The clustering results obtained with the proposed approach are more reliable than those obtained with traditional methods.

Keywords: *rock discontinuity sets, cluster analysis, orientation analysis, density peaks, decision graph, silhouette index*

1. Introduction

With the increasing demand for mineral resources and the need for special underground projects, many scholars and researchers have carried out a great deal of fruitful researches in mechanics theory of deep fractured rock mass. Discontinuity is an important component of fractured rock mass structure. Identifying rock discontinuity sets is the premise and foundation of the study on mechanical characteristics of fractured rock mass under multi-field coupling conditions (Zhou *et al.*, 2017; Zhuang *et al.*, 2014) and the study on the peridynamics of fractured rock mass (Rabczuk *et al.*, 2017; Ren *et al.*, 2017). A discontinuity is a geological interface or geologic belt formed during the geological development process in a rock mass, and it can be described by the trending direction, length and thickness. Rock discontinuities can be divided into 5 levels based on the cutting depth, extension length, fracture zone width and mechanical effects (Jia *et al.*, 2008). For engineering rock masses, grade III and above discontinuities are sparse, their distributions are regular, and their properties and boundary conditions can be explored via geophysical methods, such as engineering geological investigations, geological drilling and electromagnetic detection. Grade V discontinuities mainly affect the completeness and mechanical

properties of rock masses, and they are the main objects of statistical analyses and simulations of rock structure because of their random distribution and large quantities. Grade IV discontinuities, which mainly effect the physical and mechanical properties of rock masses, are discontinuous and observed at a small scale, and their orientation is difficult to investigate in engineering practice; thus, they are usually studied via simulation. Therefore, it is important to efficiently analyse the orientations of grade IV discontinuities.

We need to group the discontinuities into discontinuity sets before characterizing their orientations because of the regularity and directional features of their occurrence. The identification of discontinuity sets is commonly performed by using azimuthal projections, isopycnal maps or clustering analyses. The first two methods are restricted by subjective factors, and their results usually do not present clear boundaries, compactness and objectivity. In view of these restrictions, Shanley and Mahtab (1974) introduced a method based on a clustering analysis for grouping discontinuities that effectively avoids the defects of traditional methods, and many scholars have subsequently attempted to improve on their clustering algorithm for the identification of rock discontinuity sets. Jimenez-Rodriguez and Sitar (2006) proposed a method based on spectral clustering for

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grouping rock discontinuities. However, problems exist in specifying the number of clustering groups and initial clustering centres. Ma *et al.* (2015) proposed an enriched K-means clustering method for grouping fractures with meliorated initial cluster centres; however, the clustering results of this method cannot be adjusted once the classifications are generated. Based on orientations of discontinuity data, Liu *et al.* (2017) proposed a modified affinity propagation algorithm for identifying rock discontinuity sets. Cai *et al.* (2005) proposed a fusion of fuzzy C-means clustering and a genetic algorithm. These approaches avoid local minima in the calculation process but also need to specify the cluster number, clustering centres, clustering size, crossover probability, mutation probability and termination generation according to the actual situations. Song *et al.* (2016) proposed a multi-parameter method for grouping discontinuities based on the quantum particle swarm optimization algorithm, and clustering centres are determined by searching the global optimal solutions of objective functions. To eliminate the disadvantages of the dynamic clustering algorithm, Lu *et al.* (2007) proposed a method based on the niche Pareto genetic algorithm to find the optimal solution of grouping rock discontinuities. This method avoids specifying the number of clusters, although many parameters must be adjusted.

Rodriguez and Laio (2014) proposed a clustering method based on fast searches and finding density peaks called “clustering by fast search and find density peaks”. This method can effectively find the number of potential clusters and central discontinuities of corresponding clusters and does not need to specify the number of clusters and clustering centres. The discontinuities of each cluster can be divided into core discontinuities and outlier discontinuities based on the corresponding density boundaries. In this paper, we present an approach based on clustering by fast searches and finding density peaks to identify discontinuity sets. The orientation of each discontinuity is converted into pole on a stereonet by using the upper Schmidt hemisphere projection, and the sine squared value of the acute angle between each pair of discontinuities is adopted as the clustering distance which measures the similarity between discontinuities. Referring to Clustering by Fast Searches an Finding Density Peaks (CFSFDP) method, the local density and controlled distance of each discontinuity are introduced into our approach. Through calculating the decision value of each discontinuity and drawing the decision graph, the number of potential clusters and central discontinuities of corresponding clusters can be found automatically. Each noncentral discontinuity is assigned to the corresponding cluster to which the nearest discontinuity with a higher local density belongs. In every cluster, a boundary density is obtained by calculating and choosing the maximum average value of local densities of two discontinuities from the cluster and adjacent clusters. The discontinuities of each cluster are divided into core discontinuities and outlier discontinuities according to their corresponding boundary densities. The mean orientations are calculated by using the core discontinuities, thus the new approach can avoid interference from subjective factors and improve the

accuracy of identifying discontinuity sets.

2. Similarity Measures

2.1 Representation of Discontinuity Orientations

The spatial form of a discontinuity is defined as its orientation, which is normally represented by the dip direction and dip angle. Orientations of discontinuities largely determine the stability conditions of engineering rock masses, and they control the failure mechanism and influence the deformation and strength properties of rock and rock masses.

In this paper, we use the unit normal vectors of discontinuities to demonstrate the orientations in the right-handed Cartesian coordinate system (as shown in Fig. 1): The positive x-axis is horizontal to the north, the positive y-axis is horizontal to the east, the positive z-axis is vertically upward, and θ , ϕ and $\mathbf{n}$ are the dip direction, dip angle and unit normal vector of the discontinuity plane, respectively.

According to the trigonometric functions, the unit normal vector $\mathbf{n}$ can be expressed by its direction cosines $\mathbf{n} = (\alpha, \beta, \gamma)$ (Liu *et al.*, 2017; Ma *et al.*, 2015; Klose *et al.*, 2005), where:

$$\begin{cases} \alpha = \cos \theta \cos \phi \\ \beta = \cos \phi \\ \gamma = \sin \theta \sin \phi \end{cases} \quad (1)$$

2.2 Definition of Clustering Distance

Similarity between data is an important index in clustering algorithms, and two main methods are used to measure this similarity. The first method is the degree of proximity, i.e., the distance, between data, with a smaller distance corresponding to more similar data. The second method is the correlation coefficient, i.e., the Pearson correlation coefficient, of the data, with a greater correlation coefficient corresponding to more similar data. A good similarity index should maintain the resolution of the original data.

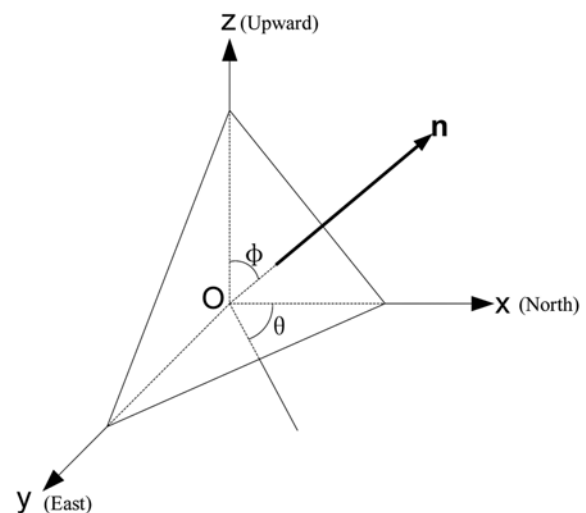


Fig. 1. Representation of the Discontinuity Orientation

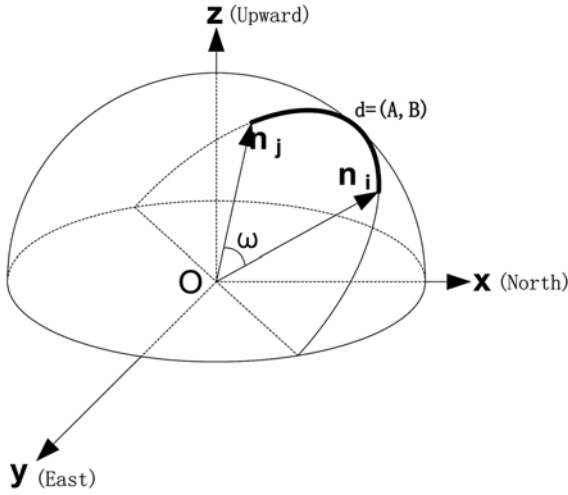


Fig. 2. Similarity Measure of Discontinuities

In engineering geological surveys, for a pair of discontinuities where the dip angles are steep and dip directions have a difference of 180° , the two discontinuities should be divided into the same set (Jimenez-Rodriguez and Sitar, 2006). If we use the Euclidean metric or spherical distance to quantify the similarity between discontinuities, the two discontinuities in the abovementioned condition will be divided into different clusters. To solve this problem, we adopt the sine-squared value of the acute angle between each pair of discontinuities as the similarity measure (Jimenez, 2008; Li *et al.*, 2015) as shown in Fig. 2. In this paper, the upper hemisphere Schmidt projection is adopted to represent the orientations of structural planes.

Let A and B represent discontinuities. θ_i and θ_j are the dip directions of A and B, respectively, and ϕ_i and ϕ_j are the dip angles of A and B, respectively. In addition, $n_i = (\alpha_i, \beta_i, \gamma_i)$ and $n_j = (\alpha_j, \beta_j, \gamma_j)$ represent the direction cosines of the unit normal vectors of the discontinuities A and B, and they can be calculated via Eq. (1).

As shown in Fig. 2, the cosine of the acute angle between two unit normal vectors expressed in Cartesian coordinates, i.e., n_i and n_j , can be calculated as:

$$\cos \omega = |n_i \cdot n_j| = |\alpha_i \alpha_j + \beta_i \beta_j + \gamma_i \gamma_j| \quad (2)$$

where the $\cdot$ symbol indicates the dot product operation.

Based on Eq. (2), the sine-squared similarity measure becomes as follows:

$$\text{dist}(A, B) = \sin^2 \omega = 1 - \cos^2 \omega \quad (3)$$

3. Methods

The principle of Clustering by Fast Searches and Finding Density Peaks (CFSFDP) is intuitive and indicates that cluster centres present two characteristics: clustering centres are the densest points within a certain range around themselves, and they are located far away from other clustering centres. The

CFSFDP method automatically finds the number of potential clusters and clustering centres. In the CFSFDP, the corresponding border region for each cluster is the point set within a certain distance from other cluster points, and the boundary density of each cluster can be calculated with the local density of the points within its border region. Outliers are thus automatically identified and excluded from the analysis, the interference of subjective factors can be avoided, and the accuracy of the clustering analysis can be improved. In contrast to the K-means method, CFSFDP does not require the number of clusters and the cluster centres to be specified. Instead, these parameters are obtained by analysing the decision graph. CFSFDP only needs to measure the distance of each pair of points and does not need to establish the objective functions or implement iterative calculations.

Existing methods for identifying rock discontinuity sets need to specify the number of clusters in advance and construct complex functions to identify optimal solutions. To solve these problems, we propose an approach based on CFSFDP for the identification of rock discontinuity sets.

We assume that data set $S = \{x_i\}_{i=1}^N$ contains discontinuity orientations, where x_i is the unit normal vector of the discontinuity i and $I_s = \{1, 2, \dots, N\}$ is the set of the corresponding discontinuity indexes. We denote the similarity between the orientations of each pair of discontinuities by $d_{ij} = \text{dist}(x_i, x_j)$, where d_{ij} can be calculated by Eqs. (2) and (3).

Considering the two characteristics of the cluster centres, we define the local density, the controlled distance and the decision value of discontinuity.

3.1 Calculation of Local Density

For each discontinuity i , its local density ρ_i (Rodriguez and Laio, 2014) is given by:

$$\rho_i = \sum_j \chi(d_{ij} - d_c) \quad (4)$$

Where,

$$\chi(d) = \begin{cases} 1, & d < 0 \\ 0, & \text{otherwise} \end{cases}$$

where d_c is a cutoff distance. From the above definition, the local density ρ_i represents the number of discontinuities that are closer than d_c to discontinuity i . When a discontinuity is farther than d_c from all other discontinuities, its local density is 0. The new approach can thus automatically and efficiently identify outliers.

To a great extent, the value of d_c affects the stability of the new approach (Bie *et al.*, 2016). If the value is too large, then the local densities of discontinuities are simultaneously large and have a low differentiating degree on the clustering process. For example, if d_c is larger than the maximum distance between each pair of discontinuities, then all local densities of discontinuities are equal and all discontinuities are divided into the same set. If d_c is smaller than the minimum distance between each pair of discontinuities, then all local densities of discontinuities are 0 and every discontinuity is divided into a different set.

The selection of d_c is based on the actual application. We can

use random data of discontinuities to benchmark the new approach under different cutoff distances to find an appropriate threshold of d_c for the identification of rock discontinuity sets.

3.2 Quantification of Controlled Distance and Decision Value

$\{q_i\}_{i=1}^N$ is the subscript set of local densities $\{\rho_i\}_{i=1}^N$ that are sorted in descending order, where:

$$\rho_{q_1} \geq \rho_{q_2} \geq \dots \geq \rho_{q_N} \quad (5)$$

The controlled distance of each discontinuity q_i (Rodriguez and Laio, 2014) can be calculated as:

$$\delta_{q_i} = \begin{cases} \min_{q_j} (d_{q_i q_j}), & i \geq 2, j < i \\ \max_{q_j} (d_{q_i q_j}), & i = 1 \end{cases} \quad (6)$$

where δ_i is the minimum distance from discontinuity i to any other discontinuity with a higher local density than discontinuity i . For the discontinuity with the highest local density, its controlled distance is the maximum distance between each pair of discontinuities.

To effectively identify the cluster centres, we draw a decision graph with decision values that are sorted in descending order of discontinuities. The decision value (Rodriguez and Laio, 2014) of discontinuity i is defined as:

$$\lambda_i = \rho_i \cdot \delta_i \quad (7)$$

If any discontinuity is one of the cluster centres, its local density ρ and controlled distance δ are larger than its neighbours, and it thus possesses a large decision value. Moreover, the discontinuities that have a low local density ρ and high controlled distance δ are regarded as outlier discontinuities. Therefore, the clustering centres of discontinuities generally show larger decision values than other discontinuities in a decision graph. We can observe a jumping point where λ_i transitions from small values to large values. In the new approach, the discontinuities with decision values larger than the jumping point are identified as clustering centres. Therefore, the clustering centres and the number of clusters can be recognized.

To quantitatively determine the jumping point, the decision values of discontinuities have been sorted in descending order in decision graph. The difference of each interval between any two adjacent points is calculated, and the interval which owns the maximum times compares to the latter interval is considered as the jumping interval.

Please note if there are several intervals own the same highest times compare to their latter interval, we select the interval which belong to the pair of points with the highest decision values. The discontinuities which above the jumping interval are regarded as the central discontinuities.

We assume the maximum number of discontinuity sets is n in an analysis. Therefore, the differences of intervals between two adjacent points in the first $n + 2$ points in the decision graph are calculated.

The difference of the number i and $i + 1$ points is as follows:

$$\Delta_{i,i+1} = \lambda_i - \lambda_{i+1} \quad (8)$$

The ratios of two differences that belong to two adjacent intervals are calculated as follows:

$$\frac{\Delta_{1,2}}{\Delta_{2,3}}, \dots, \frac{\Delta_{i,i+1}}{\Delta_{i+1,i+2}}, \dots, \frac{\Delta_{n,n+1}}{\Delta_{n+1,n+2}} \quad (9)$$

If the largest ratio is $\frac{\Delta_{i,i+1}}{\Delta_{i+1,i+2}}$, the interval between the number i

and $i + 1$ points is the jumping interval, the number i is jumping point, therefore, the first i points are central discontinuities.

Practical applications show that the number of discontinuity sets is no more than five in an identification of rock discontinuity sets, since a large number of sets could be confusing and unsuitable in actual applications. The results of analysis should be accorded with the characteristics of rock mass structures, and contain less discontinuity sets. Thus, we set $n = 5$ in the manuscript.

3.3 Cluster Boundary Density

After the central discontinuities and the number of clusters are identified, every noncentral discontinuity is assigned to the corresponding cluster to which the nearest discontinuity with a higher local density belongs. To improve the accuracy of clustering analysis, outlier discontinuities should be identified and excluded from the clustering process. The new approach divides the discontinuities of each cluster into core discontinuities and outlier discontinuities based on their corresponding boundary density.

To calculate the boundary density of a cluster, we consider the ρ_j , which is the local density of discontinuity j belonged to adjacent clusters, since the local densities of the discontinuities in the adjacent areas between any two clusters could be a reference value of determining the boundary density. In addition, considering the local densities of discontinuities from adjacent cluster could make the two clusters more distinguishable.

We assume that there is cluster set C and its boundary density is denoted by B_c , which can be calculated by:

$$B_c = \max \frac{\rho_i + \rho_j}{2}, i \in C, j \notin C \text{ and } d_{ij} < d_c \quad (10)$$

where i indicates any discontinuity that belongs to cluster set C and j indicates any discontinuity that belongs to another cluster set.

3.4 Error Detection

A potential cluster could be divided into two or multiple small clusters by mistake when the data points show a large inhomogeneity. To correct this kind of error, a detection process is introduced at the end of clustering analysis. The distance of the mean orientations for each pair of clusters is calculated in the detection process, and the two clusters that are close to each other would be merged.

A parameter ε is introduced in the detection process, the two clusters which the distance between their mean orientations is less than ε should be regarded as the same cluster. The boundary density of the merged cluster is then recalculated, and the core

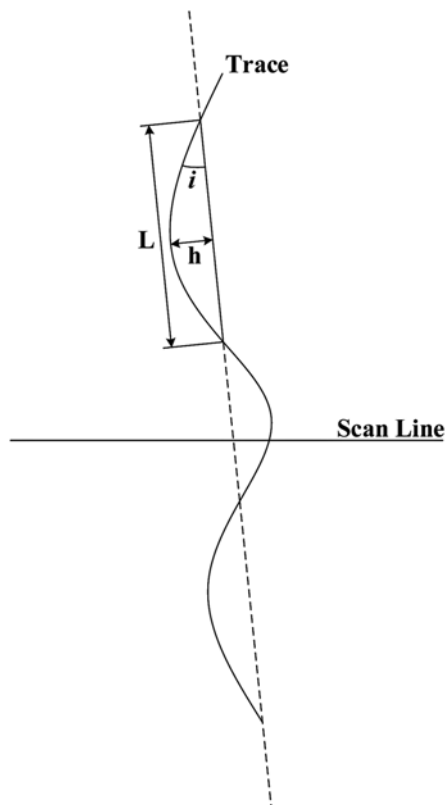


Fig. 3. Calculation Map of the Structural Surface for Undulating Angle

discontinuities, outlier discontinuities and mean orientations should be updated.

Please note that if there are more than two clusters are closer than ε , the two clusters which own a smaller distance from each other should be merged preferentially.

ε is a positive real number, its value is determined by the undulation of structural surface. In engineering geological surveys, we can estimate the undulation of discontinuities by the traces in outcrops (Chen *et al.*, 2013). The undulating angle i is used to measure the undulation of discontinuity, and it can be obtained by traces of discontinuities, as shown in the Fig. 3.

The h is average undulating height, and the L is average baseline length. The undulating angle and parameter ε can be calculated as:

$$i = \tan^{-1} \frac{2h}{L} \quad (11)$$

$$\varepsilon = \sin^2 i \quad (12)$$

3.5 General Steps

In this section, data for 30 discontinuities obtained from practical engineering are used to demonstrate the new approach. The flowchart of the new approach that explains the general process is presented in Fig. 4.

The new approach is described as follows:

Step 1 Calculate the unit normal vector of each discontinuity

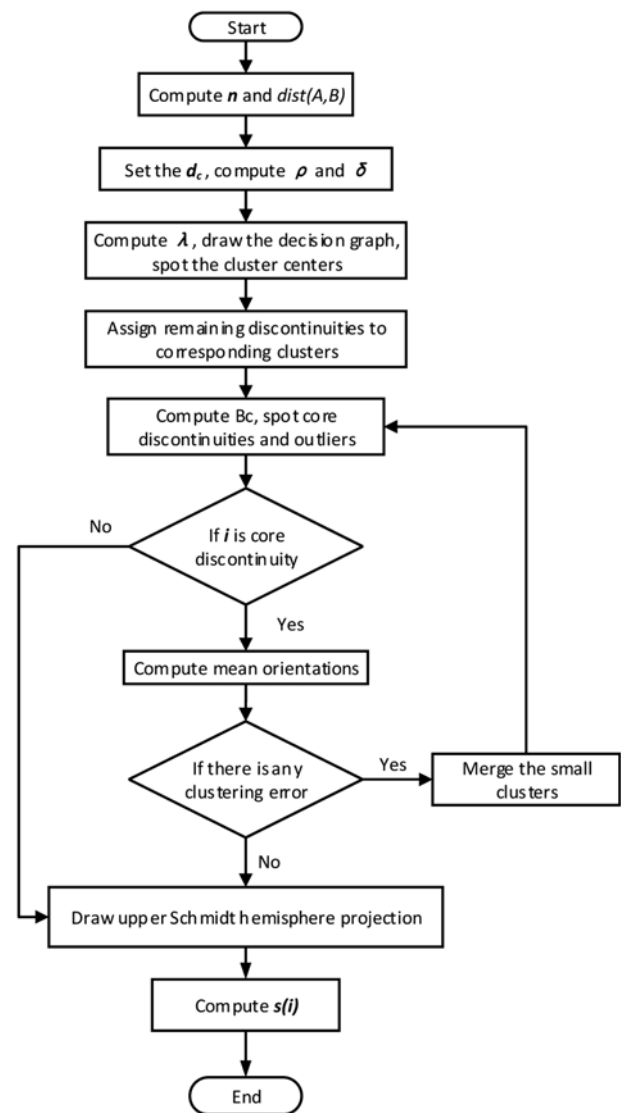


Fig. 4. Flowchart of the Fast Clustering Method for the Identification of Rock Discontinuity Sets

using Eq. (1).

- Step 2 Calculate the distance between each pair of discontinuities.
- Step 3 Set the value of the cutoff distance d_c and calculate the local density of each discontinuity using Eq. (4).
- Step 4 Sort the local densities of all discontinuities in descending order. Calculate the controlled distance of each discontinuity using Eq. (6) and identify its nearest neighbour with a higher local density.
- Step 5 Calculate the decision value λ of each discontinuity using Eq. (7). Draw the decision graph with the decision values of discontinuities sorted in descending order. Identify the discontinuities whose decision values are larger than the jumping point as cluster centres. Count the number of clusters.
- Step 6 From the larger decision value, begin to assign each noncentral discontinuity to the corresponding cluster

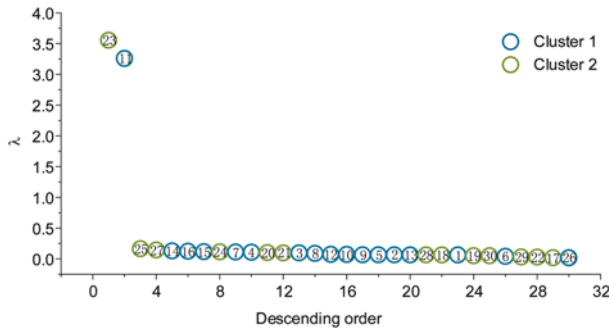


Fig. 5. Decision Graph of the Demonstration Data

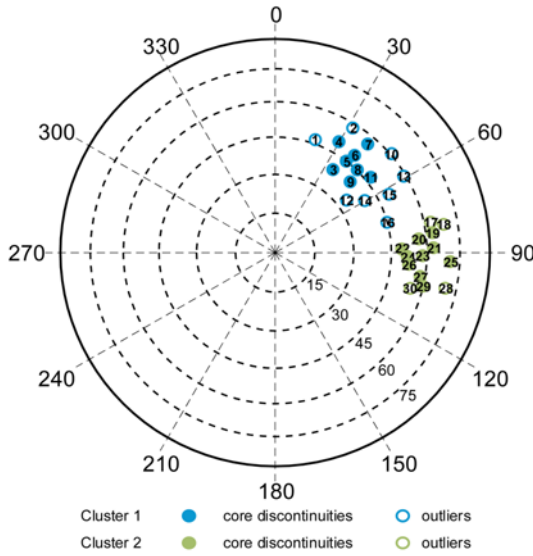


Fig. 6. Upper Schmidt Hemisphere Projection of the Demonstration Data

to which the nearest discontinuity with a higher local density belongs.

- Step 7 Calculate the boundary density of each cluster. Divide the discontinuities of each cluster into core discontinuities and outlier discontinuities according to their corresponding boundary densities.
- Step 8 Detect the clustering errors, and merge the small clusters from the same potential cluster into a correct cluster.
- Step 9 Use the upper Schmidt hemisphere projection to represent the orientations of structural planes. (See Fig. 6).
- Step 10 Calculate the silhouette coefficient to validate the clustering result.

As shown in Fig. 5, the decision values of discontinuities 11 and 23 are the maxima of all discontinuities and the remaining discontinuities are low. Therefore, we identify discontinuities 11 and 23 as the separate centres of clusters 1 and 2. Fig. 6 shows the orientations of 30 discontinuities in the upper hemisphere Schmidt projection. As anticipated, discontinuities 11 and 23 are the densest points of their corresponding clusters, and the remaining discontinuities are divided into core discontinuities

and outlier discontinuities according to their corresponding boundary densities.

3.6 Approach Validation

The effectiveness of clustering can be validated by the separation of clusters and the compactness of points in each cluster.

In accordance with the opinion proposed by Rousseeuw (1987), for any discontinuity x_i and its corresponding cluster S_i , we define d_i as the average distance between x_i and other discontinuities of S_i . Parameter d_i (Rousseeuw, 1987) can be calculated as:

$$d_i = \frac{\sum d_{ij}}{\text{card}S_i - 1}, x_i, x_j \in S_i \text{ and } i \neq j \quad (13)$$

The value of d_i measures the dissimilarity between the discontinuity x_i and cluster S_i . In extreme cases, the cluster S_i has only one discontinuity, and d_i is 0.

$Md(i)$ is the minimum average distance of x_i to all discontinuities of any other cluster (Rousseeuw, 1987):

$$Md(i) = \min \left\{ \frac{\sum d_{ij}}{\text{card}S_j} \right\}, x_i \in S_i, x_j \in S_j, S_i \cap S_j = \emptyset \quad (14)$$

where $\text{card}S_j$ is the cardinal number of cluster S_j . The value of $Md(i)$ measures the dissimilarity between discontinuity x_i and neighbouring clusters.

The silhouette coefficient (Rousseeuw, 1987) of discontinuity x_i is defined as:

$$SC(i) = \frac{Md(i) - d(i)}{\max(d(i), Md(i))} \quad (15)$$

We can easily observe the following:

$$-1 \leq SC(i) \leq 1 \quad (16)$$

for each discontinuity i .

A value of $SC(i)$ approximately equal to 1 indicates that $d_i \ll Md(i)$. A low d_i value implies that x_i is well assigned to its corresponding cluster. Larger $Md(i)$ values correspond to more accurate classifications of x_i . Therefore, a value of $SC(i)$ approximately equal to one indicates that the discontinuity x_i is well clustered. If $SC(i)$ is close to 0, then $d_i \approx Md(i)$, which means that x_i is in the boundary area of two adjacent clusters and whether x_i should be allotted to S_i or a neighbouring cluster is unclear. The unacceptable condition is that $SC(i)$ is negative, which means that x_i should be assigned to the neighbouring cluster.

The mean of all silhouette coefficients in cluster S_i can be used to evaluate the compactness of S_i . Therefore, we can take the average of the means of all clusters to validate the effectiveness of the cluster analysis.

3.7 Choice of Cutoff Distance d_c

According to Wang *et al.* (2016), the statistical behaviors of the dip direction and the dip angle are usually fit using uniform, normal or log-normal distribution functions. Based on the Monte Carlo method, we separately generate four random data sets with

Table 1. Simulation Parameters for Generating Random Discontinuity Sets

Discontinuity set #	Orientation (Bivariate normal distribution)			
	Dip direction		Dip	
	Mean	Standard deviation	Mean	Standard deviation
1	38	10	80	7
2	94	15	15	7
3	210	15	65	10
4	290	15	30	10
5	350	7	45	10

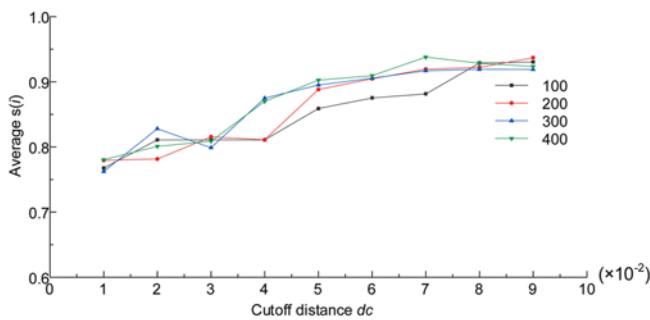


Fig. 7. Silhouette Coefficients of Random Data Sets obtained by using the New Approach to Analyse Different Cutoff Distances

quantities of 100, 200, 300 and 400. Every data set contains five clusters of discontinuity orientations that have a bivariate normal distribution, and their detailed parameters are presented in Table 1.

The cutoff distance d_c influences the local density and the decision value λ . To improve the efficiency of the new approach, an appropriate cutoff distance d_c is needed. If d_c is larger than the appropriate value, then the core discontinuities are erroneously divided into outlier discontinuities in certain disperse clusters. If d_c is smaller than the appropriate value, outlier discontinuities are not effectively identified, which will produce a large error in the cluster analysis. Fig. 7 shows the silhouette coefficients of every random data set obtained by using the new approach to analyse for different cutoff distances.

Rock mass is anisotropic material, it contains a large number of discontinuities with random distributions. The value of d_c in the new approach has a strong influence on the accuracy of clustering results. To evaluate the influence of the random parameters on the value of cutoff distance d_c , we refer to some constructive achievements about sensitivity analysis, for instance, Vu-Bac *et al.* (2016) proposed a unified approach for probabilistic sensitivity analysis of computationally expensive models, Hamdia *et al.* (2018) used three sensitivity analysis methods (Morris One-At-a-Time, PCE-Sobol' and Extended Fourier amplitude sensitivity test) to quantify the influence of input parameters on the energy conversion factor of flexoelectric materials. To determine the range of d_c in different random data sets, the three important aspects should be considered: 1) The value of the silhouette coefficient. 2) The number of outliers in each cluster. 3) The

number of clusters in the clustering results.

To find an appropriate range of d_c , we adopt three distribution functions (uniform, normal and log-normal distributions), four standard deviations (5, 10, 15 and 20) and four different sizes (120, 180, 240 and 300) of discontinuity groups in generating the random data sets.

The value of d_c is related to the degree of dispersion of the data set. Through analyzing the results of clustering the random data, the range of d_c is found to be sensitive to the standard deviation. If the data set own a larger dispersion, the corresponding appropriate value of d_c should be raised. However, the distribution functions and the number of discontinuities of the random data sets have little influence in determining the value of d_c . Normally, the value of d_c of the new approach is no more than 0.12. Based on the Eq. (3), the clustering distance 0.12 presents that the acute angle between the two unit normal vectors is approximately 20° . In other words, if $d_c = 0.12$, for any discontinuity, its local density is the total number of the discontinuities which own an acute angle smaller than 20° with the discontinuity.

A large value of d_c would decrease the differences of local densities and decision values of discontinuities, as a result, the accuracy of finding jumping point in decision graph would be declined. In addition, the local densities and boundary densities are increased by a large d_c , therefore, all discontinuities would be regarded as outliers in some clusters, since the discontinuities own a low compactness in these clusters. Plenty of clustering results of random discontinuity data show that a value of d_c larger than 0.12 in the process of clustering for identifying discontinuity sets would be inappropriate in general cases. Thus the 0.12 is regarded as the upper of the appropriate range of d_c in this paper. The authors choose the 0.005 as the lower of the range of d_c , since the value is in the bottom of the ranges of d_c of almost all random cases in the table. The recommended range of d_c in this paper is [0.005, 0.12]. However, it should be noted that the final appropriate value of d_c is determined by the actual engineering data. The authors suggest that the engineers should estimate the dispersion of the data before using the new method to identify discontinuity sets. For example, a large dispersion indicates a large value of d_c should be adopted, and a small value of d_c should be adopted when the data set shows a small dispersion.

4. Application to an Artificial Data Set

Based on engineering experience and the Monte Carlo method, an artificial data set consisting of five clusters of discontinuity orientations was generated. The detailed parameters are presented in Table 2. We used these data set to validate the new approach with a cutoff distance $d_c = 0.06$. Table 2 shows the results of the cluster analysis, and the results show that the number of core discontinuities is related to the standard deviations of discontinuity orientations. Larger standard deviations of discontinuity orientations correspond to a greater number of outlier discontinuities that are identified. The mean orientations of the core discontinuities in each cluster are consistent with the pre-set parameters. The pre-

Table 2. Comparisons between Clustering Results Obtained using the New Approach and the Data Sets Statistically defined in Advance

Discontinuity set #	Orientation (Bivariate normal distribution)				Total number	Core discontinuity		
	Dip direction		Dip			Number	Mean dip direction	Mean dip
	Mean	Standard deviation	Mean	Standard deviation				
1	38	10	80	7	50	40	37.03	79.32
2	94	15	15	7	70	69	94.16	14.11
3	210	15	65	10	50	23	207.00	63.56
4	290	15	30	10	70	34	291.35	29.49
5	350	7	45	10	60	42	350.23	43.36

*Cutoff distance $d_c = 0.06$ and mean $s(i) = 0.91$

set parameters are not the evaluation standards of clustering accuracy because outlier discontinuities in each cluster are identified and excluded from the analysis.

Figure 8 shows a jumping point where the decision values

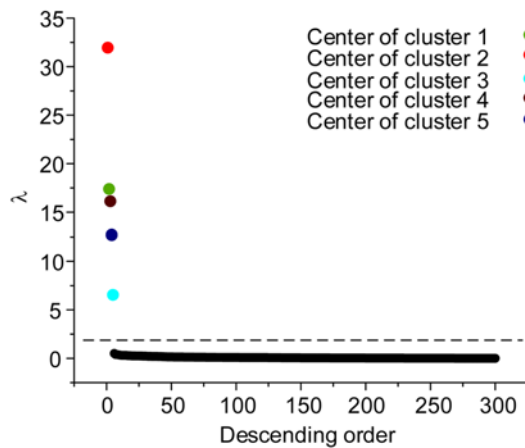


Fig. 8. Decision Graph of the Artificial Data Set

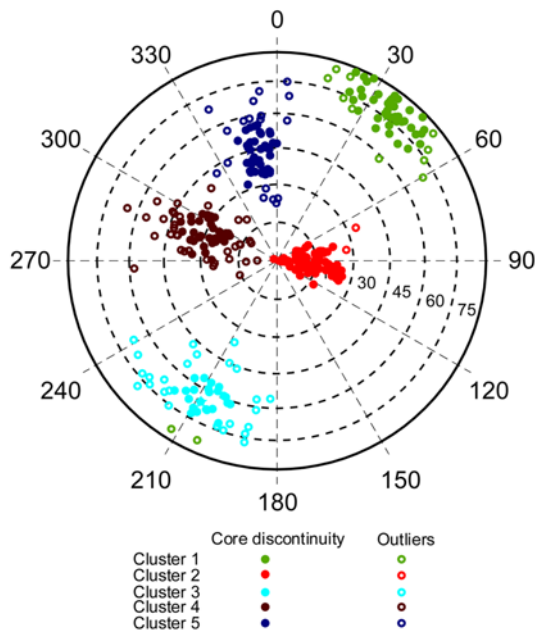


Fig. 9. Upper Schmidt Hemisphere Projection of the Artificial Data Set

transition from small values to large values. Five points are larger than the jumping point, and these 5 points are identified as the cluster centres. These results correspond to the pre-set parameters. The new approach can automatically identify and exclude outlier discontinuities from the analysis (Fig. 9), which can improve the accuracy of the clustering results. As shown in Fig. 9, in clusters 1 and 2, the new approach can automatically divide the discontinuities presenting steep dip angles and dip directions with a difference of 180° into the same cluster, and this process is consistent with engineering practices.

As shown in Table 2, the results using this new approach are similar to the statistical parameters of the artificial data set that was defined in advance, which supports the superior performance of the new approach.

5. Case Study

The Gaofeng Mine is located in Guangxi, China, and it presents

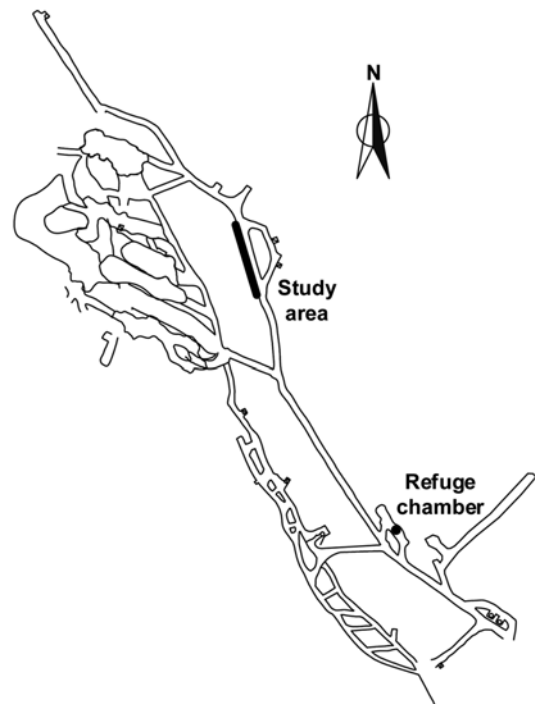


Fig. 10. Position of the Study Area in the Gaofeng Mine

Table 3. Outcrop Data of the Study Area

Serial number	Dip direction	Dip
1	60	90
2	40	56
3	35	55
4	40	65
5	60	64
6	45	60
7	25	65
...	...	...
100	46	57
101	55	58
102	70	82
103	320	70
104	46	70
105	65	65
106	65	65
107	15	56

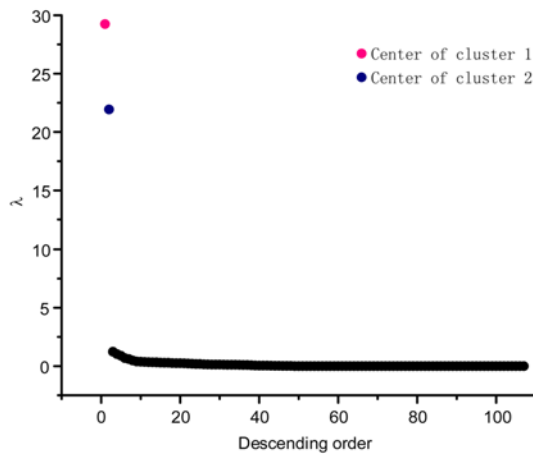


Fig. 11. Decision Graph of the Outcrop Data

two globally rare polymetallic ore shoots: the No. 100 orebody and No. 105 orebody, which is an extension of the No. 100 orebody. The No. 105 orebody concentrates at the -300 to -79 m level, and it mainly contains tin ore and concomitant lead-zinc-iron sulphide ores, and its geometric shape is an extremely complex concavo-convex lenticular mass. This orebody is hosted in middle-Devonian bioherm limestone and located in the middle section of the Da Chang anticline, and its burial depth is greater than 1,000 m, and joint fissures are developed. At present, the main middle production levels are at the -200 to -151 m level.

Figure 10 shows the position of the research area. To evaluate the stability of the rock mass at the -200 m level in the north middle section of the No. 105 orebody, we performed a discontinuity survey in the study area using the scan line method. The scan line length was 35 m, and an outcrop data set that includes 107 discontinuities was collected in the north haulage roadway out of the vein as shown in Table 3.

Figure 11 shows the decision graph of the outcrop data set

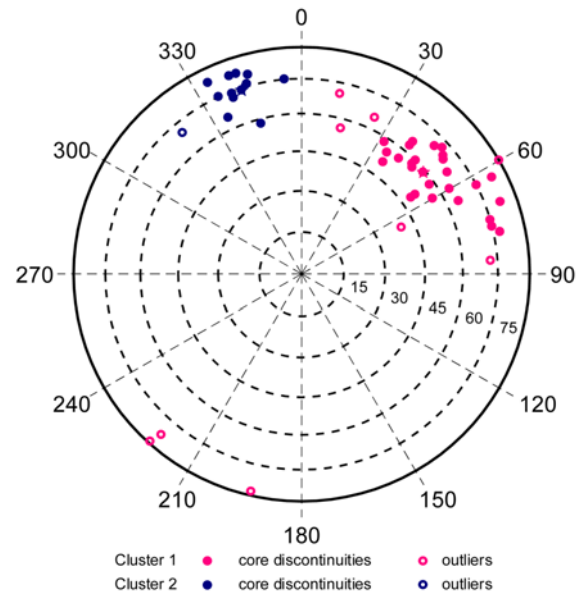


Fig. 12. Upper Schmidt Hemisphere Projection of the Outcrop Data

Table 4. Comparisons of Clustering Results Obtained using the New Approach and Rocscience Dips

Method	Cluster 1		Cluster 2	
	Dip direction	Dip	Dip direction	Dip
New approach	53.75	65.83	340.12	75.12
Rocscience Dips	53.82	65.25	339.54	75.64

*Cutoff distance $d_c = 0.09$ and mean $s(i) = 0.91$

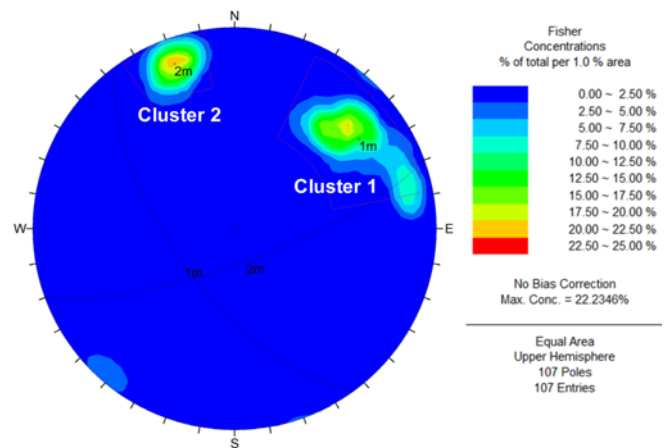


Fig. 13. Upper Schmidt Hemisphere Projection of the Outcrop Data using Rocscience Dips

using the new approach. Two cluster centres are observed in the outcrop data set, and the separate orientations of the cluster centres are $53.82/65.25$ (dip direction/dip angle) and $339.54/75.64$ (dip direction/dip angle). As shown in Fig. 12, the new approach can divide the discontinuities of the outcrop data set into two clusters and effectively identify the outlier discontinuities. In this paper, the results obtained by using the commercial software Rocscience Dips are used to validate the accuracy of the new

approach. DIPS is a geological analysis software, it contains many computational functions, for instance, statistical contouring of orientation clustering, mean orientation calculation and qualitative and quantitative feature attribute analysis. During the process of using DIPS to identify rock discontinuity sets, users draw windows which are curvilinear four-sided around data clusters on a stereonet. The poles within each window are recognized a discontinuity set, and the mean orientations of discontinuity sets are calculated. Fig. 13 shows the results using Rocscience Dips software, which indicate that two dense areas of discontinuities should be recognized as preferred orientations. Table 4 lists the results from different methods and shows that the clustering results obtained from the new approach is similar to those using Rocscience Dips.

6. Conclusions

Based on the CFSFDP method, a new approach for identifying discontinuity sets was proposed in this paper. In the new approach, the number of clusters and cluster centres does not need to be specified, and only an appropriate cutoff distance d_c is needed in the clustering process. Moreover, the outliers can be automatically identified, the interference of subjective factors can thus be avoided, and the accuracy of the clustering analysis can be improved.

The silhouette coefficient was considered an evaluation index of clustering effectiveness, and plenty of random data sets were used to test the new approach for different cutoff distances d_c . The recommended appropriate range of the cutoff distance for identifying discontinuity sets was obtained, and they ranged from [0.005, 0.12].

The performance of the new approach was evaluated using artificial data, and the results of the proposed method were consistent with statistical parameters that were specified in advance, which demonstrated the superior performance of the new approach. Moreover, the proposed approach was applied to perform discontinuity grouping in the Gaofeng Mine. The clustering results obtained using the new approach and Rocscience Dips were consistent, which further verified the reliability of the new approach.

The fractured rock mass model conformed to actual engineering can be established by using the analytic results of the new approach and the stochastic theory. Many numerical analyses of rock mass, for instance, the analysis of multi-field coupling problems of rock mass, the simulation of the seepage characteristics of fractured rock mass and the stability analysis of disturbed rock mass, can be carried out by using the fractured rock mass model.

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